

**Ex. No.: 6a**

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**FIRST COME FIRST SERVE (FCFS)**

**Aim:**

To implement First-Come First-Serve (FCFS) scheduling technique.

**Algorithm:**

1. Start the program.
2. Input the number of processes.
3. Read the burst time for each process.
4. Calculate the waiting time for each process:
  - Waiting time of process 0 is 0.
  - For others:  
 $\text{WaitingTime}[i] = \text{WaitingTime}[i-1] + \text{BurstTime}[i-1]$
5. Calculate the turnaround time for each process:  
 $\text{TurnAroundTime}[i] = \text{WaitingTime}[i] + \text{BurstTime}[i]$
6. Calculate the total and average waiting time and turnaround time.
7. Display process details, total and average times.
8. End.

**Program Code (in C):**

```
#include <stdio.h>
```

```
int main() {
```

```
    int n, i;
```

```
    int burst_time[20], waiting_time[20], turn_around_time[20];
```

```
    int total_wt = 0, total_tat = 0;
```

```
    printf("Enter the number of process:\n");    scanf("%d",
```

```
    &n);
```

```
printf("Enter the burst time of the processes:\n");
for (i = 0; i < n; i++) {
    scanf("%d", &burst_time[i]);
}

waiting_time[0] = 0;

for (i = 1; i < n; i++) {
    waiting_time[i] = waiting_time[i - 1] + burst_time[i - 1];
}

for (i = 0; i < n; i++) {
    turn_around_time[i] = waiting_time[i] +
burst_time[i];    total_wt += waiting_time[i];
total_tat += turn_around_time[i];
}

printf("Process\tBurst Time\tWaiting Time\tTurn Around Time\n");
for (i = 0; i < n; i++) {
    printf("%d\t%d\t%d\t%d\n", i, burst_time[i], waiting_time[i], turn_around_time[i]);
}

printf("Average Waiting Time is: %.1f\n", (float)total_wt / n);
printf("Average Turn Around Time is: %.1f\n", (float)total_tat / n);

return 0;
}
```

**Sample Output:**

Enter the number of process:

3

Enter the burst time of the processes:

24 3 3

| Process | Burst Time | Waiting Time | Turn Around Time |
|---------|------------|--------------|------------------|
|---------|------------|--------------|------------------|

|   |    |   |    |
|---|----|---|----|
| 0 | 24 | 0 | 24 |
|---|----|---|----|

|   |   |    |    |
|---|---|----|----|
| 1 | 3 | 24 | 27 |
|---|---|----|----|

|   |   |    |    |
|---|---|----|----|
| 2 | 3 | 27 | 30 |
|---|---|----|----|

Average Waiting Time is: 17.0

Average Turn Around Time is: 27.0

**Result:**

The FCFS Scheduling algorithm was successfully implemented. The program calculated the waiting time and turnaround time for each process and displayed the average times.